ETS2: Price Stability, Household Affordability and Environmental Integrity

A scenario-based distributional and policy-response model

Independent research project and working analytical model

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Executive summary

I developed this project to examine a central design challenge in the EU Emissions Trading System 2 (ETS2): how a credible carbon-price signal can be maintained while managing the distributional consequences for households with different levels of exposure to fossil heating and road transport. ETS2 will become fully operational in 2028 and is designed upstream, with fuel suppliers rather than households surrendering allowances. Household impacts therefore depend on the extent to which allowance costs are passed through to retail fuel prices.

The analytical model is deliberately transparent. I do not forecast the future ETS2 allowance price. Instead, I test a set of carbon-price scenarios from EUR45 to EUR150 per tonne of CO2 and apply them to three stylised household archetypes. The model combines building fuel use, road travel, direct fuel-emission factors, disposable income and cost pass-through. I then compare direct income support with structural measures that reduce ETS2-exposed fossil-fuel demand.

> *Central scenario result: at EUR75/tCO2 and 100% pass-through, the modelled annual direct carbon-cost component is about EUR162 for the vulnerable urban archetype, EUR315 for the working-household archetype and EUR512 for the rural car-dependent archetype. The purpose is to illustrate exposure differences, not to claim these are EU-average household bills.*

The core policy finding is that price containment, direct transfers and structural investment solve different problems. Direct income support can offset affordability impacts immediately but does not reduce emissions exposure. Efficiency, electrification and modal change reduce both current and future ETS2 exposure, but require capital, infrastructure and implementation time. This distinction matters when assessing the Market Stability Reserve (MSR), the Social Climate Fund (SCF) and national Social Climate Plans.

1. Research question and objectives

My primary research question is:

> *How can ETS2 manage household affordability and excessive price pressure without weakening the emissions cap or the long-run incentive to decarbonise buildings and road transport?*

I use the project to answer five practical sub-questions:

- How much direct carbon-cost exposure can different household archetypes face under alternative ETS2 prices?
- How sensitive is the result to incomplete or more-than-complete pass-through by fuel suppliers?
- Which households are most exposed when heating demand and car dependence are considered jointly?
- How do temporary transfers compare with structural measures such as energy efficiency and electrification?
- How should market-stability interventions be interpreted when lower allowance prices may also increase cumulative emissions?

2. Current policy context

ETS2 covers emissions from buildings, road transport and additional sectors. The European Commission states that it will become fully operational in 2028. It is an upstream cap-and-trade system: regulated fuel suppliers monitor emissions and surrender allowances, while households experience the policy indirectly through fuel-price pass-through. The cap is designed to reduce covered emissions by 42% by 2030 compared with 2005.

The system includes explicit safeguards for a smoother start. The Commission notes that the 2028 auction volume will be 30% higher to provide liquidity and that, during the first two operational years, additional allowances may be released from the ETS2 MSR if prices exceed EUR45/tCO2 in 2020 prices, adjusted for inflation. In June 2026, the European Parliament and Council reached a provisional agreement to strengthen these safeguards, including doubling the number of allowances that can be injected above the relevant price level and enabling earlier, more gradual releases.

The affordability architecture is not limited to allowance-supply management. The Social Climate Fund is intended to support vulnerable households, vulnerable transport users and micro-enterprises through building renovation, clean heating and cooling, zero- and low-emission mobility and, within limits, temporary direct income support. This makes it useful to compare immediate compensation with measures that permanently reduce exposure.

The Climate Desk has already analysed ETS2 MSR reform options for the German Environment Agency. Its March 2026 report highlights a key trade-off: more generous or prolonged MSR releases can

lower prices, but they increase allowance supply and cumulative emissions. My model complements that market-level question with a household-level distributional lens.

3. Model architecture

I built the model in the accompanying Excel workbook: ETS2_Household_Affordability_Model_Ruqaiya_Motagam.xlsx. The workbook contains Inputs, Household Model, Policy Scenarios, Sensitivity, Dashboard and Sources sheets.

3.1 Core household-emissions equations

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Gas emissions (tCO2/year) = Gas use (kWh/year) / 1,000 x Natural-gas emission factor (tCO2/MWh)

Petrol use (litres/year) = Road travel (km/year) x Fuel economy (L/100 km) / 100

Road emissions (tCO2/year) = Petrol use (litres/year) x Petrol emission factor (kgCO2/L) / 1,000

Total ETS2-exposed emissions = Gas emissions + Road emissions

Gross household carbon cost = Total ETS2-exposed emissions x Carbon price x Pass-through rate

Affordability burden = Gross household carbon cost / Disposable income
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3.2 Policy-response equations

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Residual emissions = Gas emissions x Gas multiplier + Road emissions x Road multiplier

Net household effect = Residual emissions x Carbon price x Pass-through - Direct transfer

Emissions reduction (%) = 1 - Residual emissions / Baseline emissions
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I keep compensation and emissions reductions separate so that a transfer is not incorrectly counted as decarbonisation.

4. Inputs and assumptions

Input	Central value	Interpretation
ETS2 operation	2028	Official policy timing
Natural-gas emission factor	0.202 tCO2/MWh	JRC/IPCC default factor
Petrol emission factor	2.28 kgCO2/litre	European Commission/JRC reporting factor
Carbon-price scenarios	EUR45, 60, 75, 100, 150/tCO2	Scenario range, not forecast
Central pass-through	100%	User-editable; sensitivity 80%-120%
Direct income support	EUR500/year	Illustrative policy input
Efficiency package	30% gas reduction; 10% road-fuel reduction	Illustrative structural scenario
Electrification	100% gas reduction; 80% road-fuel reduction	Illustrative direct fossil-fuel displacement
Deep structural package	100% gas reduction; 90% road-fuel reduction	Illustrative combined outcome

The household archetypes are intentionally stylised. They are not presented as EU averages and should be replaced with country-specific microdata for a formal distributional assessment.

Archetype	Disposable income	Gas use	Road travel	Fuel economy
Vulnerable urban household	EUR22,000	8,000 kWh/year	4,000 km/year	6.0 L/100 km
Working household	EUR35,000	12,000 kWh/year	12,000 km/year	6.5 L/100 km
Rural car-dependent household	EUR30,000	18,000 kWh/year	20,000 km/year	7.0 L/100 km

5. Worked calculation

For the working-household archetype under the central EUR75/tCO2 case:

1. Gas consumption is 12,000 kWh/year = 12 MWh/year.

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Gas emissions = 12 MWh x 0.202 tCO2/MWh = 2.424 tCO2/year
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2. Road travel is 12,000 km/year at 6.5 litres/100 km.

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Petrol use = 12,000 x 6.5 / 100 = 780 litres/year
Road emissions = 780 x 2.28 / 1,000 = 1.7784 tCO2/year
Total ETS2-exposed emissions = 2.424 + 1.7784 = 4.2024 tCO2/year
Annual carbon-cost component at EUR75/tCO2 = 4.2024 x 75 = EUR315.18/year
Monthly equivalent = EUR315.18 / 12 = EUR26.27/month
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At the assumed EUR35,000 disposable income, this carbon-cost component equals approximately 0.90% of annual disposable income. This is not the total household energy bill; it is only the direct carbon-cost component under the assumed pass-through.

6. Scenario results

Archetype	ETS2-exposed tCO2	EUR45	EUR60	EUR75	EUR100	EUR150
Vulnerable urban	2.1632	EUR97.34	EUR129.79	EUR162.24	EUR216.32	EUR324.48
Working household	4.2024	EUR189.11	EUR252.14	EUR315.18	EUR420.24	EUR630.36
Rural car-dependent	6.8280	EUR307.26	EUR409.68	EUR512.10	EUR682.80	EUR1,024.20

Three insights follow directly from the model:

- Exposure is driven by both building fuel use and mobility. A household that is moderately exposed in one category can still be highly exposed when both are combined.
- Rural and car-dependent households can face materially greater carbon-cost exposure even when their disposable income is not higher.
- The monetary effect scales almost linearly with the carbon price before behavioural response, technology switching or non-linear pass-through is introduced.

7. Pass-through sensitivity

ETS2 compliance sits upstream, so retail pass-through is a key uncertainty. For the working household, the annual carbon-cost component changes as follows:

Carbon price	80% pass-through	100% pass-through	120% pass-through
EUR45	EUR151.29	EUR189.11	EUR226.93
EUR75	EUR252.14	EUR315.18	EUR378.22
EUR100	EUR336.19	EUR420.24	EUR504.29
EUR150	EUR504.29	EUR630.36	EUR756.43

The 120% case is not a prediction that suppliers will over-recover allowance costs. It is a stress test for retail-price formation in which taxes, margins, timing and market conditions may cause observed price movements to differ from the simple allowance-cost calculation.

8. Direct compensation versus structural measures

At EUR75/tCO2 for the working-household archetype, the model compares five policy approaches:

Scenario	Gross ETS2 cost	Net household effect	Emissions effect
Baseline	EUR315.18	EUR315.18	No reduction
Efficiency & behaviour	EUR247.30	EUR247.30	Reduces gas by 30% and road fuel by 10%
Electrification	EUR26.68	EUR26.68	Eliminates direct gas use and cuts road fuel by 80%
Deep structural package	EUR13.34	EUR13.34	Eliminates direct gas use and cuts road fuel by 90%
EUR500 direct support	EUR315.18	-EUR184.82	No emissions reduction

The negative net effect in the transfer case means the illustrative EUR500 transfer exceeds this household's modelled carbon-cost component. This highlights a targeting issue: flat transfers can overcompensate low-exposure households while potentially undercompensating households with much higher exposure. Structural measures require more capital and time but reduce future exposure and emissions simultaneously.

9. Market-stability and environmental-integrity interpretation

The household model should not be used in isolation to argue for a particular MSR release rule. A lower allowance price reduces the immediate household burden, but it also weakens the price signal and may change cumulative emissions if additional allowances are injected. This is the trade-off identified in The Climate Desk's 2026 ETS2 MSR analysis.

I would therefore assess an ETS2 safeguard package against four criteria:

- Affordability: does the package prevent unacceptable short-run burdens for exposed households?
- Environmental integrity: does it preserve the cap and cumulative emissions objective?
- Investment credibility: does the carbon-price signal remain sufficiently predictable to support clean-heating and mobility investments?
- Targeting efficiency: are public funds directed toward the households and investments that reduce vulnerability rather than broadly subsidising fossil consumption?

10. Practical implementation plan for a full consulting study

To move this project from a portfolio model to a client-ready distributional study, I would complete the following workstreams:

- Household microdata: obtain country-level household budget, heating-fuel, dwelling-type, income-decile and mobility data.
- Fuel-price pass-through: estimate historical pass-through using excise/tax changes or existing carbon-pricing episodes where data permit.
- Heating technology: distinguish gas boilers, oil heating, district heat, heat pumps and electric resistance heating.
- Transport dependence: segment households by urban/rural location, vehicle ownership, commuting distance and public-transport access.
- Policy interaction: map Social Climate Plan measures, national building-renovation support, EV incentives and energy taxation.
- Behavioural response: add price elasticities for heating fuels and road transport to allow emissions and expenditure to respond endogenously.
- Distributional outputs: report impacts by income decile, dwelling tenure, geography and household type rather than only archetypes.
- Uncertainty: apply Monte Carlo or probabilistic sensitivity analysis to carbon price, pass-through, elasticities and energy demand.

11. Data request checklist

Data item	Preferred granularity	Why it matters
Household gas/heating fuel consumption	Household or income decile	Determines building exposure
Vehicle km and fuel economy	Household / vehicle	Determines road-transport exposure
Disposable income	Household / decile	Affordability metric denominator
Dwelling type and tenure	Household	Ability to invest and split-incentive risk
Urban/rural indicator	Household / municipality	Transport alternatives and heat options
Fuel prices and taxes	Monthly country data	Pass-through and retail-price decomposition
Policy support received	Household or programme	Net distributional effect
Technology characteristics	Dwelling / vehicle	Structural mitigation feasibility

12. Limitations and safeguards

- The carbon-price scenarios are not forecasts and should not be described as expected market prices.
- The archetypes are stylised and cannot substitute for representative household microdata.
- The model estimates direct carbon-cost exposure only; it does not calculate the full cost of switching to a heat pump or EV.
- Retail pass-through may vary across fuels, Member States and market structures.
- The analysis uses direct combustion factors and excludes upstream lifecycle emissions.
- The model is static unless elasticities are added; it therefore does not automatically change demand when prices rise.
- A full welfare analysis would need consumer surplus, technology-investment costs, financing constraints and policy transfers.

13. Relevance to The Climate Desk

This project directly connects to The Climate Desk's work on EU ETS2 market stability, economic modelling and policy recommendations. I use the same core policy tension - price stability versus environmental integrity - and add a transparent household-affordability model that can communicate distributional effects to policymakers. The project also demonstrates how I combine energy economics, quantitative scenario analysis and policy communication.

14. Accompanying working model

The Excel workbook is designed for transparent review and contains:

- README: model scope and boundaries.
- Inputs: all editable policy, emissions and household assumptions.
- Household Model: emissions and annual cost by carbon-price scenario.
- Policy Scenarios: compensation and structural-response comparison.
- Sensitivity: carbon price and pass-through matrix.
- Dashboard: key results and charts.
- Sources: URLs and source notes for the model inputs.

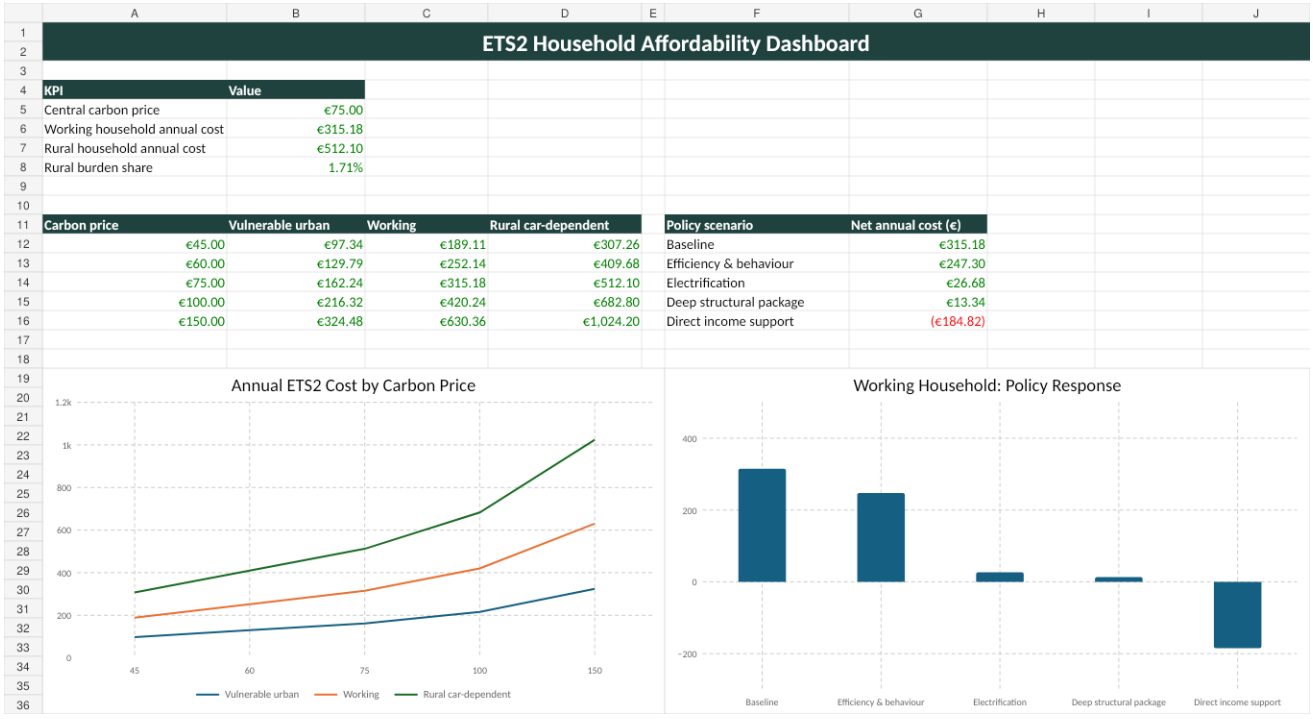


Figure 1. ETS2 model dashboard - central scenario and policy comparison.

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